

Homework 5

Nonlinear Programming

All of the problems in this homework concern equality-constrained quadratic programs.

1. Let

$$Q = \begin{pmatrix} 4 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}, \quad a = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad b = 1.$$

Verify that the matrix Q is symmetric and positive semidefinite. Then solve the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & x^T Q x + c^T x \\ \text{subject to} \quad & a^T x = b. \end{aligned}$$

2. Let Q and c be as in the previous problem. Let

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Solve the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & x^T Q x + c^T x \\ \text{subject to} \quad & Ax = b. \end{aligned}$$

Hint: Before applying the methods recently covered in class, think about what the feasible region looks like. This problem is easier than the previous one.

3. Consider a variant of the Markowitz portfolio optimization problem in which, instead of requiring that the expected return is bounded below by a constant α , we require that the expected return is exactly equal to α . Revisit the stock market data from the introduction to Markowitz portfolio optimization (the PortfolioOptimization_Complete) notebook on Sakai) and solve this problem as an equality-constrained quadratic program.

In particular, if Σ denotes the estimated covariance matrix of asset returns, r denotes the vector of estimated expected returns, and x denotes the vector of portfolio weights,

solve

$$\begin{aligned} \min_x \quad & x^T \Sigma x \\ \text{subject to} \quad & r^T x = \alpha, \\ & \mathbb{1}^T x = 1. \end{aligned}$$

Report the optimal portfolio weights and the corresponding portfolio risk, and verify that both equality constraints are satisfied.

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$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & x^T Q x + c^T x \\ \text{subject to} \quad & a^T x = b. \end{aligned}$$

$$Q^T = \begin{pmatrix} 4 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad Q_{12} = Q_{21} = 1 \quad Q_{13} = Q_{31} = 1 \quad Q_{23} = Q_{32} = 0$$
$$Q = Q^T \quad Q \text{ is symmetric}$$

$$\lambda = 0.6228, 1.7261, 4.6511 > 1 \quad Q > 0 \rightarrow Q \geq 0$$

$$x^T Q x = 4x_1^2 + 2x_2^2 + x_3^2 + 2x_1x_2 + 2x_1x_3 \geq 0$$

$$f(x) = x^T Q x + c^T x$$

$$\nabla f(x) = 2Qx + c$$

$$2Qx + c + az = 0$$

$$a^T x = 1$$

$$\begin{bmatrix} 2Q & a \\ a^T & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} -c \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 8 & 2 & 2 & 1 \\ 2 & 4 & 0 & 1 \\ 2 & 0 & 2 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -3 \\ 1 \end{bmatrix}$$

$$[x_1, x_2, x_3] = [-0.0625, 1.1875, -0.125]$$
$$z = [-2.625]$$

$$f(x^*) = x^{*T} Q x^* + c^T x^* = -0.09375$$

2. Let Q and c be as in the previous problem. Let

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Solve the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & x^T Q x + c^T x \\ \text{subject to} \quad & Ax = b. \end{aligned}$$

Hint: Before applying the methods recently covered in class, think about what the feasible region looks like. This problem is easier than the previous one.

A is an upper triangle $\det A = 1 \neq 0$

A is non singular $\text{Null}(A) = \{0\}$

$$x: Ax = 0 = \{0\}$$

$$x^* = [x_1, x_2, x_3] = [0, 0, 0]$$

$$f(x^*) = 0$$

when $\text{rank}(A) = n$, the feasible set is a single point

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Report the optimal portfolio weights and the corresponding portfolio risk, and verify that both equality constraints are satisfied.

$$A = \begin{bmatrix} r^T \\ \mathbf{1}^T \end{bmatrix} \in \mathbb{R}^{2 \times n} \quad b = \begin{bmatrix} \alpha \\ 1 \end{bmatrix}$$
$$\begin{bmatrix} 2\Sigma & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ b \end{bmatrix}$$